

Appendix 18

Density functional theory at finite temperature

The extension of DFT to finite temperature ensembles is originally due to Mermin (1965). Here we follow a slightly different path, which parallels our presentation of the zero-temperature formalism.

First of all, we work in the grand-canonical ensemble, where the hamiltonian $\hat{H}$ of Eq. (7.1) is replaced by

$$\begin{aligned}\hat{K} &= \hat{H} - \mu \hat{N} \\ &= \hat{T} + \hat{H}_{e-e} + \hat{V} - \mu \hat{N} .\end{aligned}\tag{A18.1}$$

The grand thermodynamic potential Ω of this system is given by

$$\Omega = \text{Tr} \{ \hat{\rho}_{eq} [\hat{K} + k_B T \ln \hat{\rho}_{eq}] \} ,\tag{A18.2}$$

where Tr denotes the trace of an operator, and the equilibrium density matrix is

$$\hat{\rho}_{eq} = \frac{e^{-\beta \hat{K}}}{\text{Tr}[e^{-\beta \hat{K}}]} .\tag{A18.3}$$

The equilibrium density matrix has the following variational property: the functional

$$\Omega_{V-\mu}[\hat{\rho}] \equiv \text{Tr} \{ \hat{\rho} [\hat{K} + k_B T \ln \hat{\rho}] \}\tag{A18.4}$$

is minimum when $\hat{\rho} = \hat{\rho}_{eq}$. The minimization with respect to $\hat{\rho}$ can be carried out in two steps: first, we find the minimum among the $\hat{\rho}$'s that yield a given density $n(\vec{r}) = \frac{\text{Tr}[\hat{n}(\vec{r})\hat{\rho}]}{\text{Tr}\hat{\rho}}$; second, we minimize with respect to $n(\vec{r})$. In analogy to Eq. (7.6) we define the universal functional of the density

$$F[n] = \min_{\hat{\rho} \rightarrow n(\vec{r})} \text{Tr} \{ \hat{\rho} [\hat{T} + \hat{H}_{e-e} + k_B T \ln \hat{\rho}] \} .\tag{A18.5}$$

The true grand thermodynamic potential is then given by the minimum, with respect to $n(\vec{r})$, of the functional

$$\Omega_{V-\mu}[n] = F[n] + \int [V(\vec{r}) - \mu] n(\vec{r}) d\vec{r} : \tag{A18.6}$$

this is the finite-temperature generalization of Eq. (7.5). Assuming differentiability of the functional $F[n]$, the true equilibrium density is determined by the stationarity condition

$$\frac{\delta F[n]}{\delta n(\vec{r})} = \mu - V(\vec{r}). \quad (\text{A18.7})$$

If the functional derivative exists for every “reasonable” density, then the above equation proves the existence of a unique external potential $V(\vec{r}) - \mu$ for every such density. At variance with the zero temperature case, the potential $V(\vec{r})$ is completely determined – relative to the chemical potential μ , without the possibility of adding an arbitrary constant. Furthermore, all the examples given in Section 7.2.9, of different external potentials that yield against the same ground-state, disappear at finite temperature: a thermal ensemble can never be completely “rigid” the application of an external field. Unfortunately, the resolution of these difficulties is more formal than real, as discussed in Section 7.2.8.

The finite-temperature version of the Hohenberg–Kohn theorem, namely that two different potentials $V(\vec{r}) - \mu$ and $V'(\vec{r}) - \mu'$ cannot give the same equilibrium density follows from a straightforward extension of the proof presented in Section 7.2.2. Let us assume, against the thesis of the theorem, that the two potentials give the same density n . Denoting by $\hat{\rho}$ and $\hat{\rho}'$ the two (necessarily different) equilibrium density matrices associated with the two potentials we have

$$\begin{aligned} \Omega_{V'-\mu'}[\hat{\rho}'] &= \text{Tr} \{ \hat{\rho}' [\hat{T} + \hat{H}_{e-e} + \hat{V}' - \hat{\mu}' \hat{N} + k_B T \ln \hat{\rho}'] \} \\ &= \int \{ [V'(\vec{r}) - \mu'] - [V(\vec{r}) - \mu] \} n(\vec{r}) + \Omega_{V-\mu}[\hat{\rho}] \\ &> \int \{ [V'(\vec{r}) - \mu'] - [V(\vec{r}) - \mu] \} n(\vec{r}) + \Omega_{V-\mu}[\hat{\rho}]. \end{aligned} \quad (\text{A18.8})$$

Interchanging the primed and unprimed variables, and adding the two resulting inequalities we obtain the contradiction

$$\Omega_{V-\mu} + \Omega_{V'-\mu'} > \Omega_{V-\mu} + \Omega_{V'-\mu'}, \quad (\text{A18.9})$$

proving the falsehood of the initial assumption.

The Kohn–Sham equation can also be straightforwardly generalized to finite temperatures. To this end we simply write

$$F[n] = F_0[n] + E_H[n] + E_{xc}[n], \quad (\text{A18.10})$$

where $F_0[n]$ is the non-interacting version of the F functional.¹ Then, the stationarity condition (A18.7) takes the form

$$\begin{aligned} \frac{\delta F_0[n]}{\delta n(\vec{r})} &= \mu - V(\vec{r}) - \frac{\delta E_H[n]}{\delta n(\vec{r})} - \frac{\delta E_{xc}[n]}{\delta n(\vec{r})} \\ &= \mu - V(\vec{r}) - V_H(\vec{r}) - V_{xc}(\vec{r}) \\ &\equiv \mu - V_{KS}(\vec{r}). \end{aligned} \quad (\text{A18.11})$$

The last line of this equation defines the finite-temperature Kohn–Sham potential as the potential that produces the true equilibrium density in a non-interacting system at the same temperature and chemical potential.

¹ Note that $F_0[n]$ now differs from the non-interacting kinetic energy $T_s[n]$, due to the presence of the non-interacting entropy term.

The single particle representation of the variational problem (A18.11) leads to Kohn–Sham equations in the standard form (7.25). The only differences are (i) the expression of the density in terms of Kohn–Sham orbitals becomes

$$n(\vec{r}) = \sum_{\alpha,s} f(\varepsilon_{\alpha} - \mu) |\phi_{\alpha}(\vec{r}, s)|^2, \quad (\text{A18.12})$$

where $f(\varepsilon_{\alpha} - \mu)$ is the Fermi–Dirac distribution, and (ii) the xc potential is explicitly temperature-dependent. The local density approximation retains the standard form (7.77), except that the xc energy per particle of the homogeneous electron liquid must be replaced by the temperature-dependent Gibbs free energy per particle. This has been calculated, for example, by Tanaka and Ichimaru (1989). Explicit expressions for the exchange–correlation potential at finite temperature have been given by Perrot (1982) and by Perrot and Dharma-Wardana (1984).